

DSA NOTES

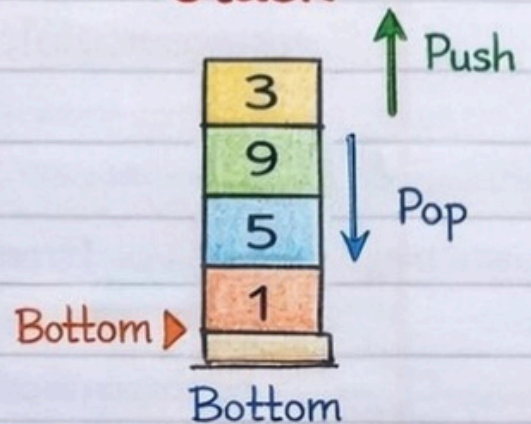
Complete DSA Notes - Part 1

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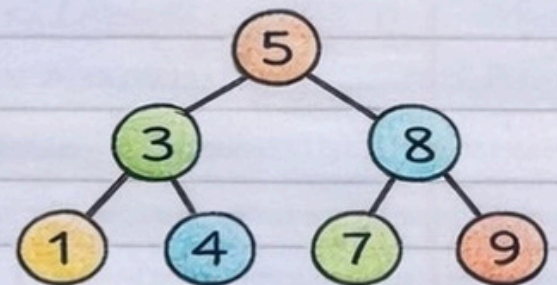
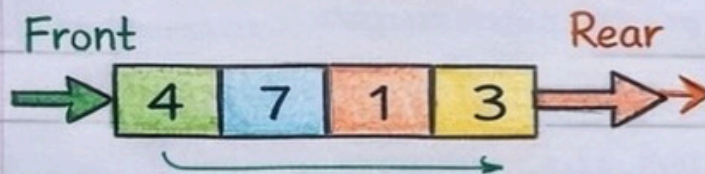
Array



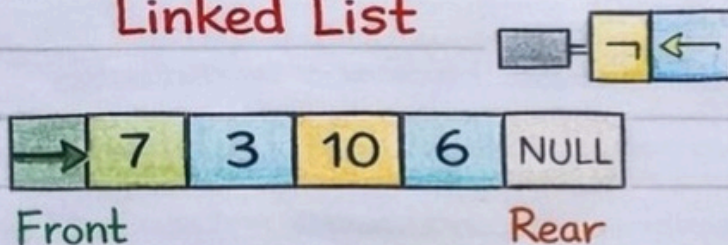
Stack



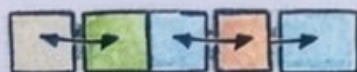
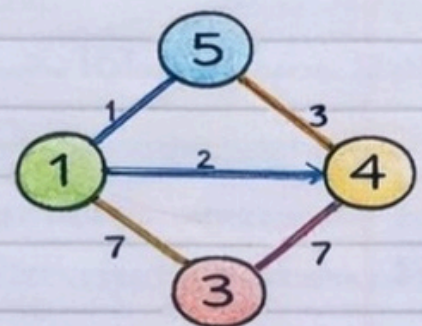
Queue



Linked List



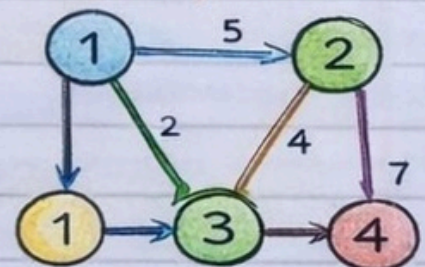
Binary Tree



Linked List



Graph

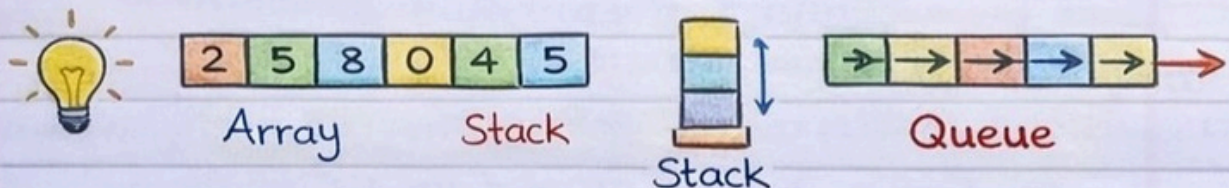


Chapter 1: Introduction to DSA

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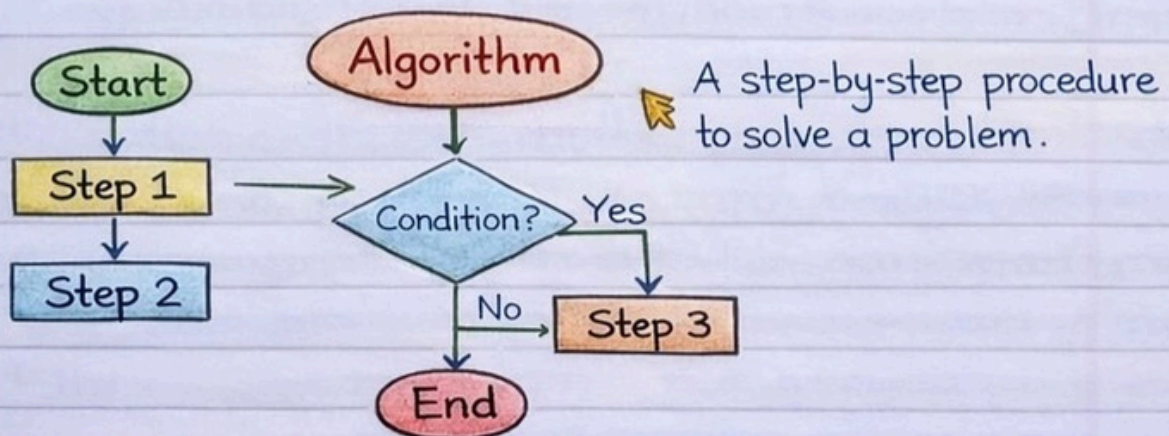
What is Data Structure?

A data structure is a way to store and organize data in a computer so that it can be used efficiently. Different data structures are designed for different kinds of data operations to make data processing more effective.



What is Algorithm?

An algorithm is a finite sequence of well-defined steps to solve a specific problem. It is like a recipe that defines the exact steps in a particular order.



Properties of Algorithms

- ✓ 1. **Finiteness** : Must terminate after a finite number of steps.
- ✓ 2. **Definiteness** : Steps must be precisely defined and unambiguous.
- ✓ 3. **Input Output** : Take zero or more inputs, produce at least one output.